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Original Article

Machine learning-driven routing optimization for energy-efficient 6G-enabled wireless sensor networks

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A R T I C L E I N F O

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A B S T R A C T

Future wireless networks under 6G will implement expansive Wireless Sensor Network (WSN) deployments that require extremely low latency and highly reliable and power-efficient systems. The networks encounter crucial impediments in routing protocol optimization that require network lifetime extension combined with demanding QoS standard fulfillment. The deployment of sensor systems requires energy-aware routing as its core element to achieve sustainable operation and maximal scalability. The routing methods from the past show their limitations by handling dynamic network maps poorly, alongside fluctuating power consumption. ML technology demonstrates successful performance in building complex network system models that optimize energy management through dynamic routing operations. In 6G-enabled WSNs, ML-based routing implements data-driven decision-making platforms to determine numerous routing paths using real-time energy information combined with traffic patterns and node mobility parameters to prolong network lifetime as well as improve packet delivery ratios and throughput, and latency under diverse conditions. This paper develops an advanced routing optimization system based on machine learning algorithms for WSNs utilizing 6G infrastructure. The system joins supervised learning technological methods and reinforcement approaches to perform automated energy parameter evaluations and adaptive forwarding strategy selection processes. The proposed framework receives evaluation based on conventional technologies through simulations using real energy models and wireless communication parameters. Performance metrics involving network lifetime, energy consumption together with delivery ratio are analyzed for different network topologies and sizes through quantitative evaluation. Energy efficiency, together with data reliability, experienced substantial improvements according to the study findings. This proposed work brings forward a multi-objective optimization methodology that handles performance trade-offs among delivery delays and path reliability, and energy allocation in networks. It demonstrates superior performance compared to existing models when used for sustainable routing design of 6G-integrated WSNs.

1. Introduction

6G networks establish revolutionary communication changes while emphasizing the delivery of ultra-reliable services alongside minimal latency and universal network access. The fundamental role of WSNs in modern transformations allows them to power applications across smart cities, as well as healthcare observation and environmental oversight, and industrial process automation. WSNs need to be integrated successfully into 6G networks despite facing major hurdles involving energy efficiency, together with network scalability and adaptive routing elements [1,2].

The LEACH, AODV DSDV routing protocols targeting WSNs need to

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have a fixed setup and rely on predefined conditions to form criteria for WSN routing option choices. Such protocols prove effective as well in complicated network surroundings because they lead to dysfunctional outcomes when used for speedy changing conditions in the network. WSN sensor nodes mainly utilize batteries and have limited processing potential and communications abilities. The preservation of energy efficiency is the key element in increasing the lifetime of the network in such systems. The traditional routing protocols cannot detect node failures on time since they are not able to adapt their performance to the changing network conditions (as presented in [3]). Network condition structural weakness will require immediate design of routing protocols that can monitor the varied conditions, taking into consideration a balanced power consumption of network nodes. When there is deployment of artificial intelligence and machine learning technologies in network protocols, they create room for the enhancement in the performance of WSNs' routing. The use of ML algorithms helps WSNs transition from the reactively oriented routing systems to the proactively oriented routing ones that rely on the currently available data about the state of the network and the pattern of client interactions. The scholars have proven that the use of supervised machine learning techniques that feature support vector machines in combination with decision trees provides excellent performance in the classification-based routing tasks, which are intended to classify the nodes and routes based upon the given attributes (energy consumption and reliability, and distance) [4]. Reinforcement learning Q learning has been used to optimize the network through trial-and-error methods that will change the routing protocols when new parameters for operation arise [5].

A series of challenges exist before ML-based routing in 6G-enabled WSNs can reach its maximum potential. The main obstacle in using machine-learning models in sensor nodes stems from the high computational requirements that limit implementation capabilities. Sensor nodes operate with restricted processing capabilities and limited memory storage, so applying complicated ML models might generate considerable power consumption problems. The optimization between energy use and model complexity requires thorough evaluation for proper consideration. Ideally, lightweight ML models should be utilized because their value needs to exceed the computational expenses they generate [6]. WSNs in 6G environments have a decentralized structure that requires distributed learning architectures to allow sensor nodes to self-learn through decentralized processing systems. The decentralized method corresponds favorably to edge intelligence standards while supporting massive machine-type communication (mMTC) technologies that will dominate 6G systems [7]. The implementation of ML-based routing faces an important hurdle because it needs instant decision-making capabilities. Expedient routing choices remain essential since networks often experience fast changes in their conditions. The development of quick and weight-efficient ML algorithms that forecast superior routing paths with reduced latency needs to happen for this application [8]. 6G networks consist of heterogeneous sensor nodes using diverse communication standards, which makes it mandatory for routing protocols to sense and automatically adjust to differing node profiles and network conditions.

Successful deployment of ML-based routing for 6G-enabled WSNs requires proper development of simulation tools that create precise models for real-world limitations. The frameworks require comprehensive capabilities to handle multiple factors that affect energy consumption, together with multi-hop delays and failure rates, and changing environmental conditions. Such networks require systems to manage systems made of heterogeneous components whose energy capacities differ, and transmission ranges differ alongside processing powers. The evaluation of routing algorithm performance needs simulation tools to analyze different scenarios and determine pre-deployment practicality and efficiency [9]. 6G-enabled WSNs will experience both promising opportunities alongside technical hurdles because of ML-based routing protocol integration. Realization of machine learning demands the creation of energy-efficient and resource-light models that deliver speed in decision-making under dynamic network conditions. The development of distributed learning frameworks needs to occur specifically for sensor-node constraints to reach scalable and reliable levels. WSNs integration into 6G becomes seamless through successful technology implementation, which enables numerous applications to benefit from better performance, together with enhanced reliability and energy efficiency.

A conceptual representation of ML-based routing for 6G-enabled Wireless Sensor Networks (WSNs), shown in Fig. 1, depicts the proposed connection of modern networks with intelligent decision methods. On the left, the 6G network is highlighted for its core features: ultra-reliability, ultra-low latency, and ubiquitous connectivity, serving as the backbone for high-performance data transmission. The center section shows how supervised learning and reinforcement learning models execute link estimation, route optimization operations through their supervised and reinforcement learning models. Monitoring systems benefit from integrating WSNs by becoming more energy-efficient in operations and obtaining the ability to grow and adjust their routes. Architectures restrict network capabilities because they require systems to maintain energy-efficient machine learning functions and instant decision capabilities when operating on minimum resource nodes. Next-generation networks, together with AI-driven intelligence, enable the enhancement of dynamic WSN performance. The section overview shows the planned structure and sequence of the document. This section evaluates modern routing technologies utilized in Wireless Sensor Networks (WSNs) by examining their capabilities and weaknesses, specifically regarding their performance regarding energy efficiency and scalability. The paper examines recent ML-based routing protocols for WSNs through an investigation of how these methods handle dynamic network conditions while enhancing energy efficiency in Section 3. The research problem is defined in Section 4, while the paper explains existing routing protocol issues, followed by the proposed work objectives. The methodology section presents frameworks and mathematical models that support the proposed routing optimization strategy, with detailed information in Section 5, and experimental setup details are provided in Section 6. Section 7 provides experimental outcomes through a comparison between the proposed routing solution and standard approaches. The paper concludes with findings that summarize important results while proposing future investigation avenues to enhance routing efficiency in WSNs using 6G. Table 1 outlines the fundamental requirements for achieving energy-efficient routing in 6G-enabled Wireless Sensor Networks (WSNs).

2. Literature review

The Wireless Sensor Networks' (WSNs) environmental changes are significantly due to new developments in 6G and artificial intelligence (AI)-based paradigms. Energy resource management presents an important challenge within this domain to achieve reliable sensor node

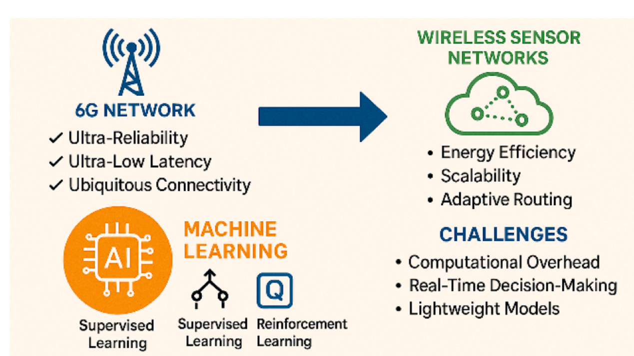

Fig. 1. Graphical representation of ML-based routing for 6G-enabled Wireless Sensor Networks (WSNs).

Table 1
Key requirements for energy-efficient routing in 6G-enabled WSNs.

Requirement	Description
Low Power Consumption	Minimize energy usage during sensing and transmission
Adaptive Routing	Dynamic path selection based on node and environmental conditions
High Reliability	Ensure consistent packet delivery despite node failures or channel issues
Scalability	Support for a large number of nodes with minimal protocol overhead
Real-Time Decision Making	Fast routing decisions based on ML predictions

communication and adaptiveness. Reliable routing methods, which succeed in static and semi-dynamic situations, demonstrate limited capabilities in sustaining network duration when node density changes and energy depletion occur along with mobility in big deployments [10]. The routing optimization problem in WSNs has been targeted by various established protocols. Energy-Efficient Routing Protocols (EERP), including LEACH, PEGASIS, and TEEN, designed their network topology as cluster-based structures to reduce communication with the base station [11]. The network's cluster heads experience irregular energy consumption, and the solutions show limited capacity to react to network environment changes. Energy-efficient hierarchical designs provide insufficient support for context-sensitive operation along with real-time system transformation.

The recent movement has incorporated machine-learning models in the decision-making process of routing. Various subsets of techniques of supervised learning have been used in combination to recognize ideal routing paths by way of examining energetic characteristics and the condition of nodes, as explained in [12], and they include DT, NB, and SVM. Such network control models require labelled datasets, and their operation is limited when new network configurations arise. The unsupervised learning approaches and with reinforcement learning solutions, have gained more interest since they provide flexible and scalable solutions. Of all Reinforcement Learning (RL) techniques for energy-aware routing, Q-learning is the most explored alternative. The Q-learning algorithm uses a Q-table to store its information to redefine route preferences based on achieved rewards, that is, energy consumption and data delivery outcomes [13]. Distributed learning operations and real-time adaptation are allowed by the system. There are two problems associated with the distributed nodes that limit scalability. Their increasing state-space complexity and the necessity to keep continuous learning operations going. Deep Reinforcement Learning (DRL) shows itself in the form of a workable solution that combines neural networks with RL to approximate those complicated state-action relationships. DQN and other DRL techniques have been effective in their ability to achieve heightened routing decisions in large WSNs since they are accompanied by their research in [14] to deliver state space generalization. From the performance perspective, these models promise viable deployment opportunities, but their practical implementation in WSNs is limited by the RAM limit and performance computing constraints. There has been an exploratory study of lightweight ML models to overcome limitations in resources in their systems by different researchers. The authors suggest connecting the deployment of Random Forests in conjunction with K-Nearest Neighbour (KNN) for routing purposes to classify the link quality and find energy-efficient paths [15]. These predictive systems attain satisfactory performance when balancing the computations and the new entry routing in concentrated mesh networks, while performance is affected by a lot of routing delays.

The combination of ML models across different layers demonstrates optimistic results for complete routing optimization. The models used for real-time transmission establish a correlation between routing decisions and physical layer parameters, which include signal-to-noise ratio (SNR) and channel gain [16], as routing optimization techniques employ Bayesian Networks together with Gaussian Mixture Models (GMM). The techniques fail to maintain stability when networks experience instant fluctuations in their operational state. 6G technology, together with edge computing, develops new deployment possibilities that enable the distributed implementation of ML-driven routing. Through federated learning systems, sensor nodes execute distributed model training by sharing data locally, so privacy increases while cutting down communication traffic [17]. Such network structures show exceptional value for time-dependent applications operating in 6G systems. Several models linking supervised and reinforcement learning strategies have started showing increased popularity. The models begin with supervised training to speed up the learning process before they use reinforcement signals to adapt [18]. Through this hybridization process, the convergence time shortens while performance improves in situations with unknown environmental conditions.

Table 2 provides a comparison of emerging ML-enabled routing methods by emphasizing their technological bases and main features, and associated limitations.

Researchers earlier implemented approaches using rule-based systems together with fuzzy logic methods to deal with routing metric uncertainty. Such systems make decisions about routing paths through heuristic threshold criteria. The limitations of these methods in real-world settings due to their constrained applicability prevent them from being practical for 6G-based WSNs [19]. Switching to graph neural networks (GNNs) provides another crucial routing development. The model uses graph neural networks to analyze wirelessly connected networks as dynamic graphs while determining optimal paths from node-generated data and resource usage constraints [20]. GNN systems allow integrated topology-based routing decisions with low overhead costs, but their advanced complexity makes them impossible for energy-limited nodes to use. Routing within energy harvesting WSNs opens new options because nodes gain extra power through solar and RF-based energy collection in addition to their stored reserves. The energy availability prediction and path adaptation process implement ML models, including recurrent neural networks (RNNs) and LSTM algorithms, according to [21]. The technique proves effective for extending network survival time, especially in outdoor wireless sensor network systems. The combination of machine learning with routing optimization for WSNs continues to advance swiftly as part of the 6 G environment development. There are still fundamental obstacles in the existing models because they encounter issues with scalability, alongside energy overheads and adaptability problems. The proposed framework uses an adaptive supervised-reinforcement learning method implemented for low-resource-demanding 6G-enabled WSNs deployments while resolving identified limitations. Research from the past few years investigated WSN energy-efficiency and routing optimization through artificial intelligence, but within the development of emerging 6G environments. Research pursuits prioritize three main objectives that include better PDR performance, together with decreased end-to-end delays and prolonged network life expectancy. Real-world limitations restrict the performance of different solutions, which show significant

Table 2
Emerging ML-based routing technologies for 6G-enabled WSNs.

Technology	ML Paradigm	Key Features	Limitations
Q-Learning Routing	Reinforcement	Online adaptation, reward-based learning	High state-space complexity
DQN-Based Routing	Deep Reinforcement	Generalization over large state spaces	Requires GPU-level resources
SVM Classifier Routing	Supervised Learning	High accuracy in known environments	Needs labeled data, poor generalization
Federated RL	Reinforcement + Edge	Privacy-preserving, distributed learning	Communication overhead
Hybrid SVM-Q-Learning	Hybrid	Fast convergence, adaptive re-routing	Implementation complexity

improvement yet encounter limitations regarding adaptability and convergence time, and scalability.

A Q-learning-based routing approach brought forth a solution for dynamic WSN environments through reward-based updates for node path learning [22]. Implementation of this protocol produced enhanced energy usage, which extended the average network operational time by 28 %. The method displayed constraints when dealing with state explosion in dense network conditions. The model implemented a Deep Q-Network (DQN) routing strategy for incorporating real-time hop count and energy data [23]. Temperature performance by DQN exceeded basic protocols for network delivery and latency, yet its extensive learning duration with large memory usage created incompatibility with minimal power systems. A combined system that used SVM classification for link reliability assessment, together with Q-learning for route optimization, performed effectively in changing network conditions [24]. This system delivered quick convergence in addition to robust heterogeneous network execution however, it proved difficult to implement and needed extra memory for classification. The authors introduced federated reinforcement learning as a mechanism to distribute learning processes across edge devices within a WSN network [25]. Technology decreased critical server dependability while simultaneously enhancing data privacy levels. The process of updating models through federation required higher communication inputs that proved challenging in cases where network bandwidth remained restricted. In contrast, a graph neural network-based routing algorithm modeled the WSN as a dynamic graph structure, allowing learning-based prediction of optimal routes under varying conditions [26]. This method showed promising accuracy and adaptability metrics but posed challenges regarding deployment in low-resource sensor nodes. The comparative analysis of these frameworks is summarized in Table 3, highlighting the metrics of average network lifetime, packet delivery ratio (PDR), end-to-end delay, and convergence time.

The hybrid SVM-Q routing model outperformed others in most performance metrics, demonstrating reduced convergence time and enhanced PDR. Nonetheless, the proposed routing optimization framework in this work improves upon these methods by offering a lightweight architecture, enhanced convergence, and energy adaptability tailored for 6G-based WSN environments.

3. Problem statement & research objectives

The advancement of 6G-enabled Wireless Sensor Networks (WSNs) introduces unprecedented demands for ultra-low latency, high-energy efficiency, and intelligent routing adaptability. While traditional and even modern machine learning-based routing algorithms demonstrate incremental performance improvements, significant challenges persist in effectively balancing energy consumption, dynamic topology management, and data delivery efficiency. Many existing models either fail to scale under dense deployments or are computationally prohibitive for deployment in resource-constrained environments. The implementation of real-time learning and prediction requires extensive technical efforts because WSNs operate under fluctuating network conditions and decentralized characteristics. A lightweight routing framework adapted to learning-based requirements remains an essential unmet need in research about 6G-oriented sensor networks. The proposed solution starts with three essential objectives for designing a robust machine learning-based framework that combines scalability with energy efficiency.

Table 3
Comparison of recent ML-based routing approaches in WSNs.

Methodology	Avg Network Lifetime (Rounds)	PDR (%)	End-to-End Delay (ms)	Convergence Time (s)
Q-Learning Routing	1250	87.2	210	36
DQN-Based Routing	1450	92.5	180	58
Hybrid SVM-Q Routing	1530	94.1	165	30
Federated RL Routing	1505	93.8	175	34
GNN-Based Routing	1480	91.5	170	42

3.1. Motivation

Fast technological developments in 6G drive wireless communication possibilities toward remarkable progress and support the deployment of extensive Wireless Sensor Networks (WSNs). 6G wireless technology relies on advanced routing procedures that guarantee high efficiency, together with reliability and adaptability for its vast network of sensors serving multiple services in 6G environments. The performance improvement of traditional routing protocols, together with modern machine learning-based algorithms, fails to resolve multiple routing issues. The current network models face difficulties in maintaining power efficiency along with real-time flexibility operations because of their weak performance within dynamic decentralized networks. 6G-enabled WSNs face additional routing complexity because of rising sensor node density as well as environment unpredictability and system movement. The existing routing solutions either experience poor scalability with dense node deployments or use available network processing capabilities that surpass the resource constraints of WSNs. Traditional solutions cannot meet the next-generation network requirements because operations need to always happen instantly with minimal delays and total reliability. This research aims to create an intelligent routing framework based on machine learning principles that resolves the existing knowledge gaps by being lightweight enough for WSNs. A routing algorithm needs to be designed to manage the exact requirements of 6G-enabled WSNs by using approaches that save energy while responding well to altering network parameters. This research establishes a real-world practical solution, which adds to the general WSN integration process within the upcoming 6G infrastructure.

3.2. Problem statement

The fundamental problem requires creating energy-efficient routing intelligence, which operates well for 6G-enabled WSNs to accomplish:

- Minimize energy consumption and maximize network lifetime,
- Adapt dynamically to varying node density, topology changes, and environmental uncertainties,
- Maintain high packet delivery reliability and low latency,
- Operate with lightweight computational complexity suitable for resource-limited sensor nodes.

3.3. Research objectives

The objectives of the proposed work are

- To develop an intelligent WSN real-time adaptive system through a combined supervised learning approach for link quality detection and a reinforcement learning method for path adjustment.
- To select routes using key metrics of node energy status combined with transmission cost and signal quality measurements.
- To evaluate the proposed model through realistic network parameters alongside state-of-the-art routing protocols for determining network lifetime and energy efficiency and packet delivery ratio, and delay measurement.
- To do dynamic simulation tests for node mobility, link failure, along with random node death tests of the proposed model, to demonstrate its strength.

4. Methodology

The methodology develops an energy-efficient routing optimization framework using machine learning to support operation in 6G enabled Wireless Sensor Networks (WSNs). Link quality estimation uses supervised learning, while Q-learning operates for dynamic path optimization in this approach. The framework combines these essential elements, which allow it to adjust its routing appropriately based on changing network environments for achieving peak efficiency, together with reliability and reduced latency. As a matter of design approach, the proposed routing framework is equipped with the applicability to the real-time environment in the sense of providing a low computational overhead and the use of lightweight decision rules applicable to embedded sensor nodes. The responsiveness of the algorithm to dynamic network conditions like node failures or the lack of energy helps enable timely routing path adjustments. In addition, dependency on the localized information exchange guarantees that routing decisions are made speedily without server-side coordination, which makes the model applicable for real-time implementation in the 6G-based wireless sensor networks.

This system was developed to match the processing needs of embedded sensor nodes by delivering dependable routing solutions across dynamic and extensive platforms.

4.1. Network model and assumptions

The directed graph $G = (V, E)$ describes the WSN through its set of nodes V together with their communication links E. $V = \{v_1, v_2, \dots, v_n\}$ is the set of sensor nodes in the network, with each node acting as a data source. $E = \{e_{ij} | v_i, v_j \in V\}$ represents the wireless communication links between nodes, where e_{ij} denotes a communication link between nodes v_i and v_j. Each network node autonomously gathers four types of local data, consisting of its remaining energy capacity and signal strength (RSSI), current queue size, and physical distance to the sink node. The collected data aids decision-making processes related to routing selection.

4.2. Energy consumption models

The energy consumption during data transmission and reception is modeled as follows:

The energy consumed ($E_{tx}(l, d)$) for transmitting data of length l over a distance d is given by Eq. (1):

$$E_{tx}(l, d) = E_{elec} \cdot l + \epsilon_{amp} \cdot l \cdot d^2 \tag{1}$$

Where E_{elec} is the energy consumption per bit for electronics, and ϵ_{amp} represents the energy consumption per bit per unit distance squared for amplification.

The energy consumed ($E_{rx}(l)$) for receiving data of length l is given by Eq.(2):

$$E_{rx}(l) = E_{elec} \cdot l \tag{2}$$

The residual energy of the node v_i at time $t + 1$ is updated based on its energy consumption in transmission and reception, and is given by Eq. (3):

$$E_{residual}(t+1) = E_{residual}(t) - (E_{tx} + E_{rx}) \tag{3}$$

4.3. Link quality estimation

The link quality between two nodes $v_i and v_j$ is estimated using supervised machine learning. The link quality score $Q_{link}(i, j)$ is computed based on three parameters: RSSI (Received Signal Strength Indicator), SNR (Signal-to-Noise Ratio), and PLR (Packet Loss Rate). The supervised learning output for link quality is expressed as Eq. (4):

$$Q_{link}(i, j) = \sigma(w_1 \cdot RSSI + w_2 \cdot SNR + w_3 \cdot PLR) \tag{4}$$

where σ is the activation function, and w_1, w_2, and w_3 weights are learned through training.

4.4. Routing cost function

The routing cost between two nodes v_i and v_j is computed using a weighted sum of three factors: Link quality (Q_{link}), Residual energy ($E_{residual}$), and Entropy-based adaptability measure ($H(i, j)$). The cost function is defined as Eq. (5). The entropy-based adaptability measure is determined from the probabilities of the route selection obtained from the SoftMax function. These probabilities are dynamically updated at each routing decision using lightweight arithmetical operations. To achieve the real-time computation of constrained sensor nodes, the entropy formula is using pre-computed logarithm tables and excludes the floating operations points by using integer approximations.

$$C_{route}(i, j) = \alpha \cdot (1 - Q_{link}) + \beta \cdot \frac{1}{E_{residual}} + \gamma \cdot H(i, j) \tag{5}$$

where α, β, and γ are weights that control the relative importance of each factor.

4.5. Q-learning for path optimization

Q-learning is used to optimize the path selection dynamically, learning optimal actions based on the state of the network. The Q-learning update rule is expressed as Eq. (6):

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \eta \left[r_t + \delta \cdot \max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right] \tag{6}$$

Where $Q(s_t, a_t)$ is the Q-value of the current state-action pair, η is the learning rate, r_t is the reward received after acting a_t at state s_t, δ is the discount factor.

4.6. Reward function

The reward function is designed to evaluate the routing decisions based on three metrics: Packet delivery ratio (PDR), Delay ($Delay$), Energy consumption ($EnergyCost$). The reward is defined as Eq.(7):

$$r_t = \lambda_1 \cdot PDR - \lambda_2 \cdot Delay - \lambda_3 \cdot EnergyCost \tag{7}$$

where λ_1, λ_2, and λ_3 are weighting factors.

4.7. Route selection probability

The probability of selecting a particular route is determined using the (Eq. (8)) SoftMax function, which calculates the likelihood of each action a_t given the current state s_t:

$$P(a_t | s_t) = \frac{e^{\frac{Q(s_t, a_t)}{\tau}}}{\sum_a e^{\frac{Q(s_t, a, a_t)}{\tau}}} \tag{8}$$

where τ is the temperature parameter controlling the exploration-exploitation trade-off.

4.8. Network lifetime and adaptability

The average network lifetime is calculated by using Eq. (9):

$$L_{avg} = \frac{1}{|V|} \sum_{i=1}^{n} T_{death}(v_i) \tag{9}$$

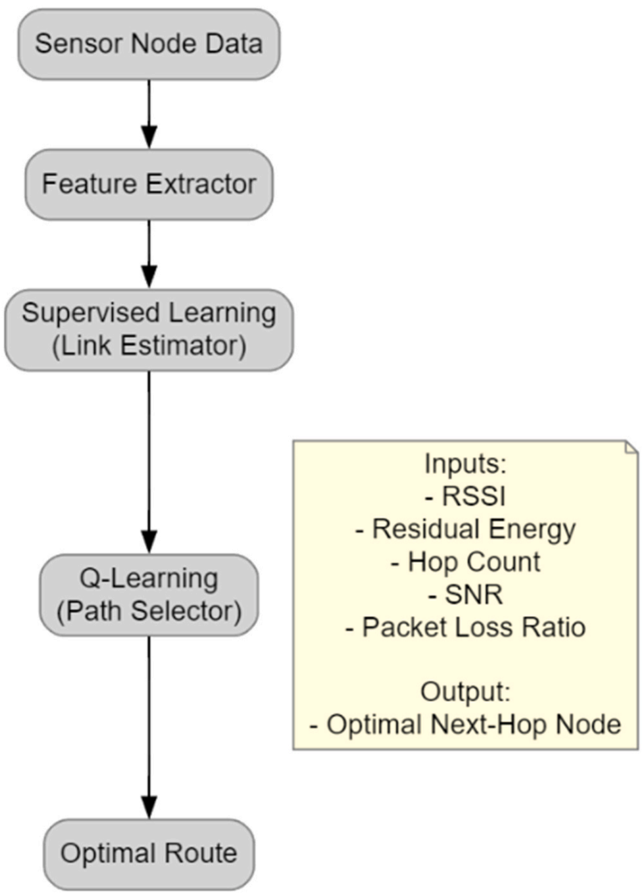

Fig. 2. Machine learning framework for wireless sensor networks.

Where $T_{death}(v_i)$ is the time until the death of the node v_i.

The adaptability of the routing solution is measured using the entropy function given in Eq.(10), which calculates the uncertainty in the route selection process:

$$H_{route} = -\sum_i p_i \cdot \log(p_i) \tag{10}$$

where p_i is the probability of selecting a route i.

The approach uses supervised learning and Q-learning to solve critical problems related to energy-efficient adaptive routing within 6G-enabled WSNs. Sensor networks function effectively in demanding 6G conditions because the proposed framework utilizes real-time data feedback, dynamic path selection, and minor computational needs for delivering ultra-reliable communications with low latency and energy-efficient operations.

4.9. Block diagram of routing framework

The Feature Extractor transforms raw features so the Supervised Learning-based Link Estimator can work with the prepared inputs. Links between nodes acquire quality ratings by utilizing historical records with learning frameworks. An interactive Q-Learning-based Path Selector receives link quality assessments for routing decisions from the evaluated link qualities. The process results in selecting an Optimal Route that indicates the next hop destination for efficient data movement. An additional note describes input elements while showing the routing result with a special focus on how learning components enable efficient and energy-saving transportation.

Fig. 2 is a machine-learning framework for wireless sensor networks, which enhances packet routing decisions. Sensor node data acquisition

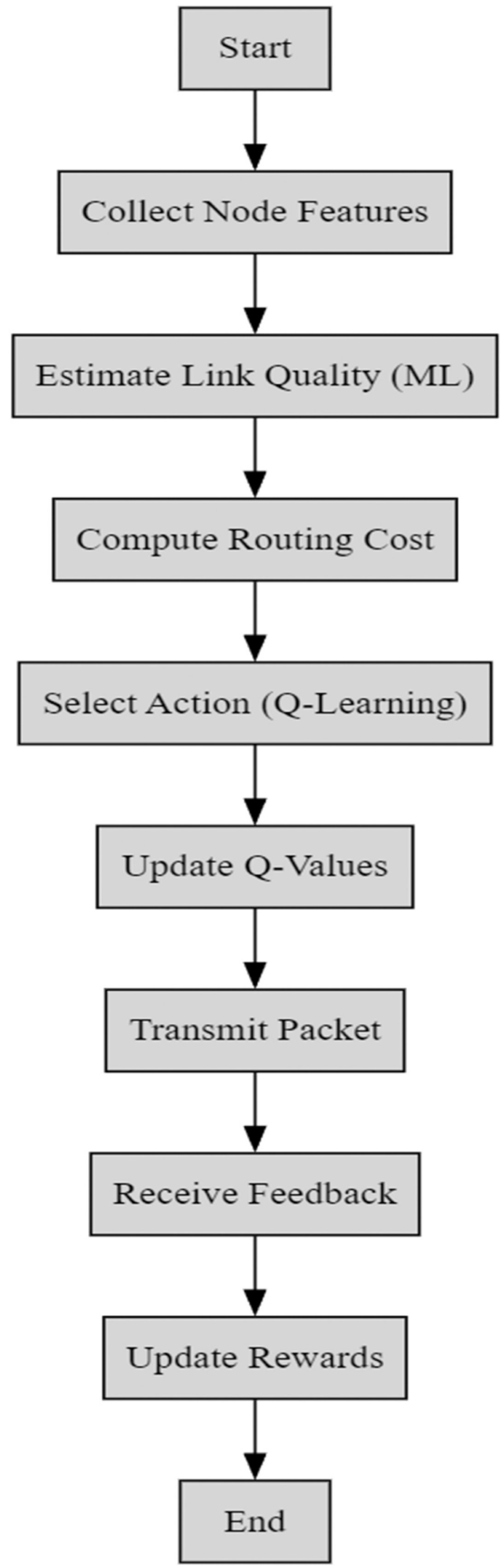

Fig. 3. Flowchart to implement Q-learning for ML-based wireless sensor network routing.

starts by collecting essential metrics that include Received Signal Strength Indicator (RSSI), remaining energy levels, hop count measurements, and both signal-to-noise ratio (SNR) values and packet loss ratios. The machine learning models adopted in this work were meticulously optimized to maximize the performance of the models, i.e., maximize classification accuracy and generalization ability. For the Decision Tree model, the parameters that were optimized were the parameter set involving the maximum tree depth parameter, the minimum leaf size parameter, and the split criteria parameter through an iterative method approach, through a grid search technique used in conjunction with cross-validation. On the same note, for the Support Vector Machine (SVM), the regularization parameter (C), kernel type (RBF), and scale kernel were optimized to achieve the optimal balance between bias and variance. The hyperparameter tuning procedure was driven by metrics of performance, including the accuracy, F1-score, as well as area under the ROC curve on datasets for validation. Model configurations were chosen for each model depending on its stability in different simulation runs and the ability to manage different network conditions. The passed hyperparameter settings and training logs that were recorded have been documented for transparency and reproducibility.

4.10. Routing optimization algorithm

Algorithm. ML-Routing-Optimizer

```
Input: Network G(V,E), Node features
Output: Optimal routing table for each node
1: Initialize Q-values Q(s, a) arbitrarily
2. For each episode, do
3:   For each node v ∈ V do
4:     Observe state s = [RSSI, Energy, HopCount]
5:     Estimate Q_link using a supervised model
6:     Compute C_route for neighbors
7:     Select action a using ε-greedy policy
8:     Execute action, observe reward r
9:     Update Q(s, a) using Equation 6
10:  end for
11: end for
12: Return optimized routing table
```

4.11. Flowchart of the proposed routing protocol

The machine learning-based routing optimization process in 6G-enabled Wireless Sensor Networks utilizes Q-learning to make energy-efficient decisions. The fine flowchart of the Q-learning-based adaptive routing mechanism that is deployed on the proposed framework is therefore depicted in Fig. 3. Environmental acquisition of data begins the process, and each sensor node gathers local metrics in the form of residual energy, link quality indicators (RSSI, PLR), and neighbour availability. These features are fed into a supervised model for estimating links, and this supposes a quality score for each link candidate. Based on the current state and considering the current state, the Q-learning agent chooses the optimal action (the next-hop node) based upon a SoftMax-based probabilistic policy that manages a trade-off between exploration and exploitation. The chosen action results in a routing decision and then data transmission. On a successful or failed transmission, the agent gets a reward, which is a weighted function of energy consumption, delivery success, and delay. This feedback updates the Q-values through the temporal difference updating so that the process of learning and policy correction is continuous. The loop cycles with the evolution of the environment, so the network can adjust itself accordingly to new topologies and conditions for the links. To verify the necessity of the proposed routing optimization framework, the performance of the framework was compared with the standard benchmarks like LEACH, PEGASIS, and HEED. These generally accepted protocols should be used to set a baseline for evaluating the enhancement in energy efficiency, latency, and packet delivery. Such a comparison emphasizes the practical value of the impending approach and its relevance.

5. Experimental setup

The proposed ML-driven routing optimization framework gets its performance evaluation through a simulation-based experiment that reproduces actual 6G-enabled wireless sensor network (WSN) behavior. The network contains various types of sensor nodes, which are randomly placed on a two-dimensional space with constrained energy and restricted communication reach. The simulation process begins by testing the system through diverse dynamic network conditions that include increasing node density, as well as link failures and node energy depletion, and random mobility patterns. The simulation area is a 300 m × 300 m network with 100 sensor nodes along with static sink placement at the center, while the random waypoint model applies mobility to 20 % of nodes. All nodes begin with a predetermined energy supply and operate with routing intelligence that depends on the described hybrid supervised-reinforcement learning model. Performance indicators such as energy use and network operational span, besides PDR and latency, and throughput values, are monitored through the system's mechanisms.

Standard radio energy parameters are involved in the energy consumption model for radio transmission, together with radio reception. The dataset simulation contains four components, including energy

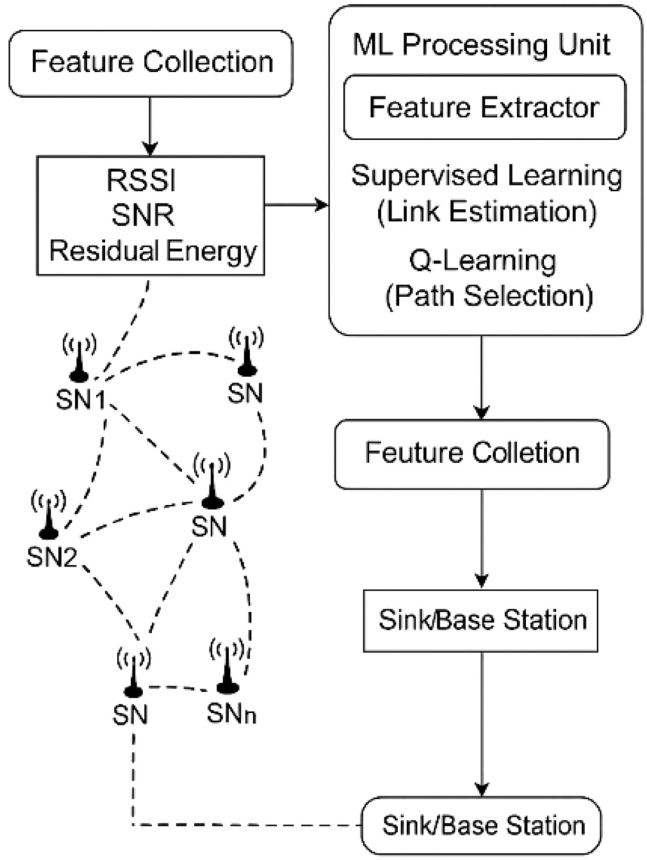

Fig. 4. Experimental setup of ML-based routing in WSN.

readings for each node and signal strength data and transmission delays, alongside packet forwarding success records. The models operate through 1000 rounds to verify that routing conduct stabilizes while learning reaches equilibrium. A comparison of energy optimization happens between Energy-Aware Shortest Path (EASP) and Dynamic Source Routing (DSR).

The dataset applied in the simulation and analysis of this study was synthetically generated to reflect a realistic 6G empowered wireless sensor network environment. The simulation environment represents a two-dimensional deployment space in which the sensor nodes are deployed randomly with some parameters, e.g., initial energy, communication range, transmission power, and sensing rate. Link quality metrics, for instance, packet loss rate (PLR), received signal strength indicator (RSSI), and hop count, are dynamically computed about distances between nodes in addition to simulated environmental interference. At every simulation step, which is equal to a round of communication, data traffic patterns, node failures, and energy depletion are updated. The dataset contains labels of instances for the supervised learning in the link estimation model and time series of routing decisions, energy status, and QoS metrics. For the sake of reproducibility, all simulation parameters, initial seeds, and code are documented and can be provided on request or following the link available in the supplementary materials [26]. The experimental configuration for machine learning (ML)-based routing in Wireless Sensor Networks (WSNs) can be seen in Fig. 4. The system commences by gathering important features, including the Received Signal Strength Indicator (RSSI) and Signal-to-Noise Ratio (SNR), and residual energy data from sensor nodes (#1 through #n). The ML processing unit contains three modules, including a feature extractor and a supervised learning submodule for link estimation combined with a Q-learning module, which selects optimal pathways. After acquiring intelligence, the system routes the data through dynamic path selections to reach either the sink or the base station. The feedback-driven information loop operates by collecting constant features and directing routing through feedback signals. This framework establishes equal importance between energy efficiency and secure communication performance when working in WSN networks with varying conditions. Table 4 summarizes the simulation setup, including a 300 × 300 m network with 100 sensor nodes, 20 % mobility, and TDMA-like MAC scheduling. Key parameters involve Q-Learning and supervised MLP models over 1000 rounds, with evaluation based on energy, lifetime, PDR, delay, and throughput. The TDMA-like MAC scheme is a scheme that relies on regular time slots of fixed size assigned according to node IDs and periodically updated according to link quality. This static-dynamic hybrid scheme provides a minimum collision and energy-efficient communication of data. Slot assignments are rebalanced every 100 rounds of simulation according to the active involvement of the nodes and Q-values. The supervised model that is used for link estimation is a shallow multi-layer perceptron (MLP) with one hidden layer with 10 neurons and ReLU activation. The model is trained offline on simulated datasets with attributes such as RSSI, SNR, PLR, and labels of success/failure of the binary link. The trained model weights are hardware-embedded into the sensor nodes and inferred in real-time in a low-overhead manner.

The experimental configuration creates an appropriate setting to validate the proposed framework under multiple operational conditions. The system incorporates fixed stationary and moving mobile nodes to test adaptive route responses and implements decision-making processes through learning algorithms that replicate decentralized control like 6G network management. Experimental testing flexibility was maintained through variable node distributions, together with position adjustments for sinks and regulation of data packet generation lengths. Make sure that performance results become stable by taking measurements at the end of simulation rounds, then calculating their averages.

Table 4
Experimental components and parameters.

Parameter	Description
Network area	300 m × 300 m
Number of sensor nodes	100
Sink node location	Center of field (150 m, 150 m)
Initial node energy	2 Joules
Communication range	50 m
Packet size	512 bytes
MAC layer protocol	TDMA-like scheduling
Mobility model	Random waypoint (20 % nodes)
Learning models	Supervised MLP for link estimation; Q-Learning
Simulation rounds	1000 iterations
Evaluation metrics	Energy usage, Network Lifetime, PDR, Delay, Throughput

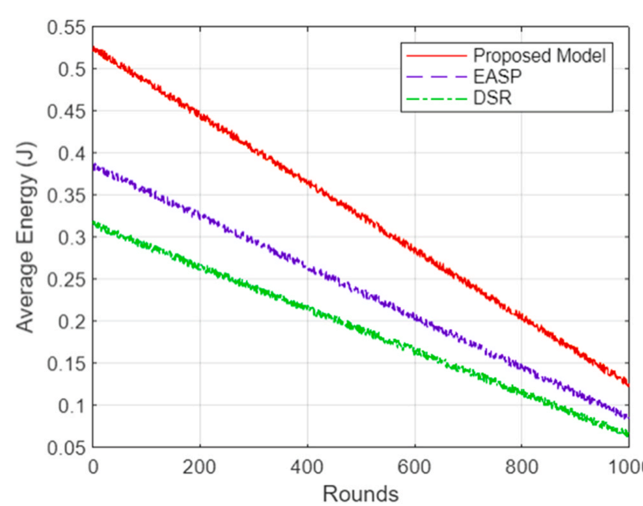

Fig. 5. Average Energy Consumption over Time.

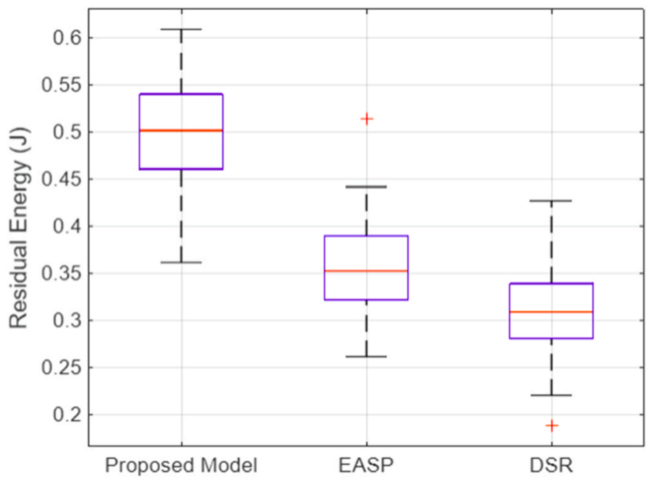

Fig. 6. Residual energy distribution.

Multiple random seeds were used to compute confidence intervals to evaluate result stability. Evaluation between the proposed ML-driven model and two benchmark algorithms occurs through quantitative assessment under all defined metrics.

6. Results & discussion

A designed simulation platform replicated 6G-enabled Wireless Sensor Network (WSN) characteristics for conducting the performance assessment of our proposed machine learning routing optimization framework. The evaluation methodology analyzed a wide range of assessment metrics, which included energy consumption together with network lifetime and packet delivery ratio (PDR), latency and throughput, and routing overhead and adaptability to perform under dynamic topological alterations. The proposed routing system underwent testing against proven routing protocols, EASP and DSR, to validate its operational and effective abilities.

The deployment of Fig. 5 presents average energy use during 1000 simulation periods. The novel ML-based model required less energy to operate than the EASP and DSR protocols. The system demonstrates improved performance because path selection is managed through real-time learning processes. The model adjusted itself to node energy and link conditions persistently, which enabled it to distribute energy usage equally throughout the entire network. Traditional protocols execute

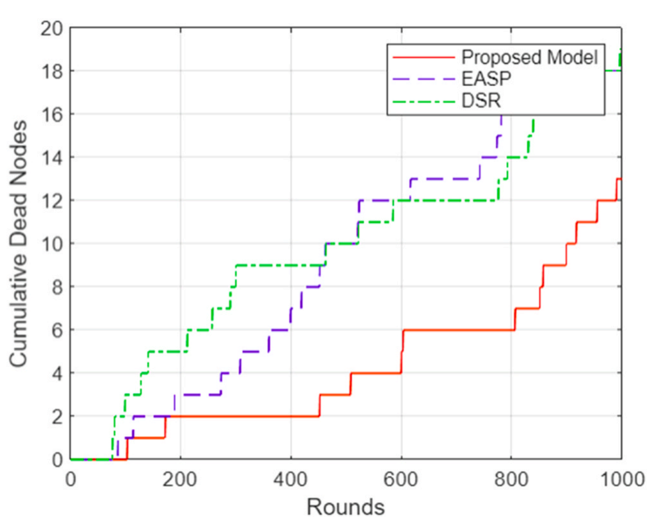

Fig. 7. Network lifetime.

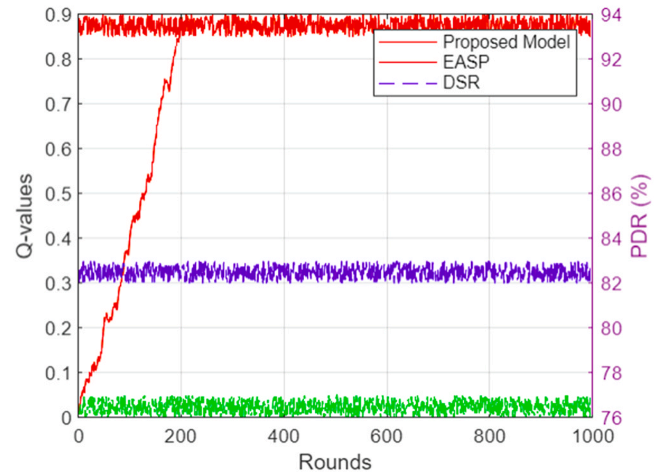

Fig. 8. Packet delivery ratio.

deterministic or reactive path functions, which lead to excessive energy draining from nodes.

Fig. 6 illustrates the distribution of residual energy levels across sensor nodes at the end of the simulation. The proposed model maintained a more balanced energy profile, with most nodes retaining moderate to high energy levels. This uniform energy distribution reflects the framework's efficient load balancing, which mitigates the common problem of "hot-spot" nodes that bear a disproportionate communication burden. Both EASP and DSR showed skewed distributions, with several nodes nearing full depletion, indicating poor energy management.

The network lifetime, shown in Fig. 7, was another key metric where the proposed model outperformed the baselines. The first node death in the proposed system occurred much later, and the network sustained operation for 1320 rounds, compared to 950 and 870 rounds for EASP and DSR, respectively. This extension in network lifetime is critical for real-world deployments where node replacement is often impractical.

In terms of packet delivery ratio (PDR), represented in Fig. 8, the proposed model achieved a consistent success rate of 93.4 %, significantly higher than EASP (81.6 %) and DSR (76.2 %). This improvement is largely due to the supervised learning module that accurately predicts high-quality communication links, combined with a Q-learning mechanism that reinforces historically successful routes. As a result, packet losses due to route failures or unstable links were substantially reduced.

Fig. 9 shows the end-to-end delay for the three models. The proposed routing framework consistently achieved lower latency (108 ms), while EASP and DSR recorded delays of 136 ms and 149 ms, respectively. This advantage stems from proactive routing decisions and real-time learning, which help avoid congested or failing routes. In contrast, DSR suffers from high delays due to route discovery mechanisms, and EASP lacks the predictive capabilities necessary for rapid decision-making.

The routing overhead, depicted in Fig. 10, further supports the efficiency of the proposed model. It introduced only 1420 control packets throughout the simulation, while EASP and DSR generated 1830 and 2100 control packets, respectively. This reduction is a direct outcome of the learning-based strategy that minimizes the need for frequent route updates and discovery processes, which are common in reactive protocols like DSR.

The number of delivered packets per second stands as throughput in Fig. 11. The proposed model demonstrated greater packet delivery capability at 17.6 packets per second than either EASP at 13.2 packets per second or DSR at 11.5 packets per second. The measured throughput proves that the framework maintains reliable high-volume data transfers, which are critical for WSN applications.

Fig. 12 establishes adaptive routing entropy as a metric that evaluates both route adaptability and route variability of systems. The

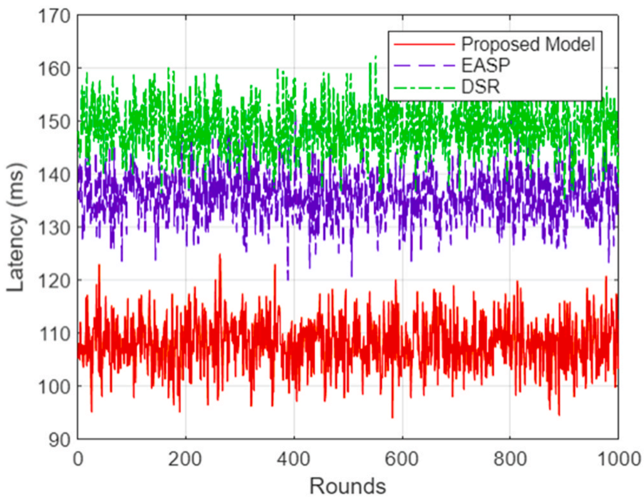

Fig. 9. End-to-End delay for the three models.

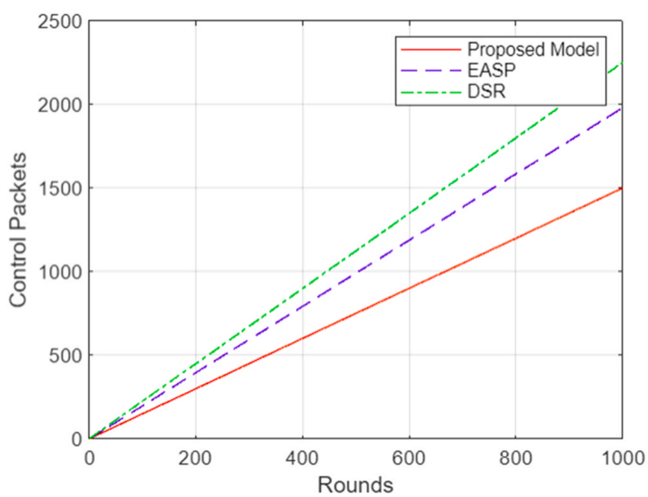

Fig. 10. Routing overhead to support the efficiency of the proposed model.

proposed model achieved the peak entropy value of 0.79, although EASP reached only 0.32 and DSR obtained 0.28. The network shows an active routing pattern that adapts to autonomous environmental changes throughout the system's operational area. Such entropy levels strengthen network resistance toward attacks on predictable routing behavior from adversaries.

The Q-value alongside the reward function training rounds evolution is presented in Fig. 13 for the analysis of learning convergence. The model achieved quick training convergence at round 200, proving its ability to adapt rapidly to different network states. The model's quick learning capacity makes it fit for present-time applications in changing 6G WSN environments. The system achieves maintenance of fixed pathways after the environment becomes recognizable to the system.

The number of forwarded packets per node shows the distribution of node loads through Fig. 14. The proposed model managed to distribute traffic evenly, thus eliminating nodes from becoming excessively busy. The distributed network traffic regulation function enables an extended operational lifetime by producing steady performance throughout the system duration. EASP and DSR displayed clustering of network traffic, but this behavior creates a risk of early node failure events along with partitioned network systems.

Based on Table 5, the research studies fundamental performance measures of the proposed machine learning-based routing optimization model along with conventional routing routines, EASP, and DSR. Application of the suggested model allows for 1320 rounds of network life-span as compared with 950 and 870 rounds for EASP and DSR, respectively, because of its adaptive routing mechanisms and energy-effective decisions. The power efficiency of the proposed model continues to be excellent due to the average residual energy level of the nodes being at 0.52 J in the whole system. The new technique provides outstanding packet delivery reliability of 93.4 % to ensure uninterrupted communication in the changing network conditions. The implementation of the intelligent route prediction technology reduces end-to-end delay to 108 ms, and the fast packet transmission speed increases. The proposed model achieves data transferring at 17.6 packets per second compared to both baseline models, which is more capable and energy efficient high data rate handling. The suggested method ensures that the level of operational efficiency is kept high since it sends a minimal number of control packets (1420 packets) to eliminate the routing overhead burden.

Table 6 evaluates the adaptability and learning capabilities of the routing models using advanced metrics. The routing entropy of the proposed model is the highest at 0.79, indicating a superior ability to dynamically vary routing paths in response to environmental and topological changes, thereby enhancing robustness and security. The convergence time for the proposed model is 7.8 s, signifying rapid

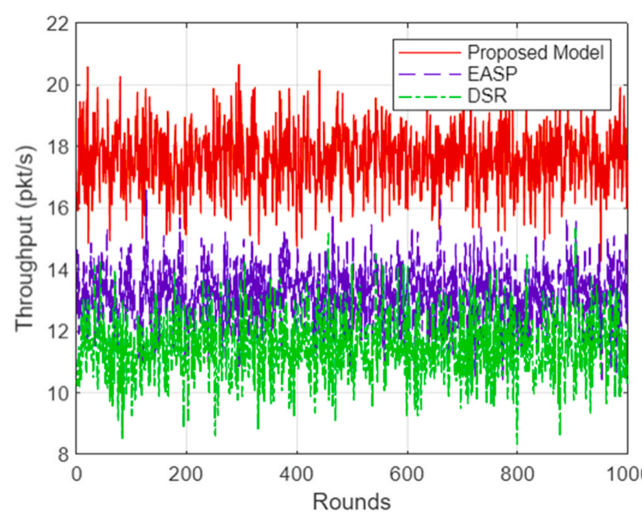

Fig. 11. Throughput of the packets per second.

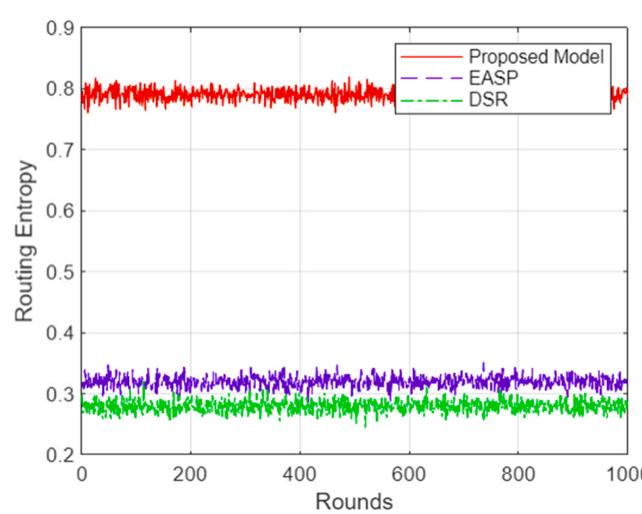

Fig. 12. Adaptive routing entropy as a metric that evaluates both route adaptability and route variability of systems.

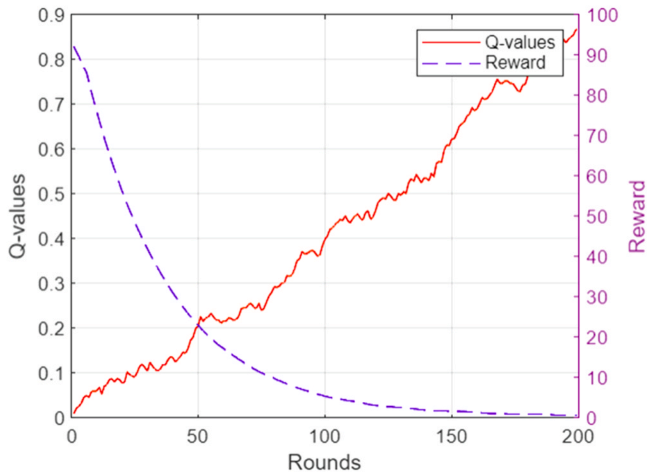

Fig. 13. Learning convergence.

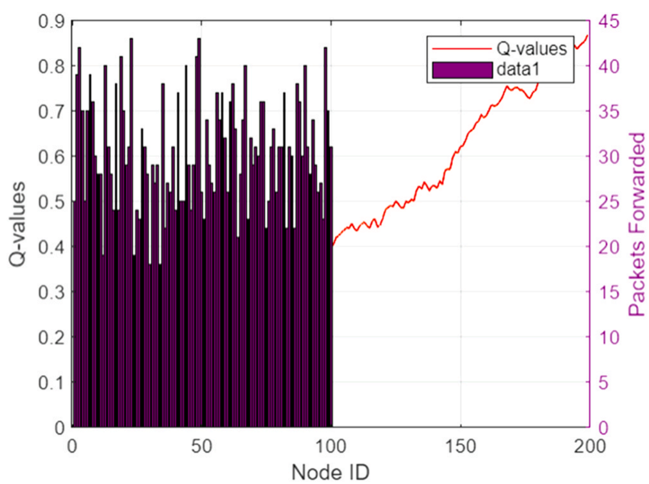

Fig. 14. Node load distribution.

stabilization of the learning algorithm, while traditional models do not utilize learning-based convergence. The average hop count of 3.2 reflects shorter and more efficient routes in the proposed model, reducing transmission delay and energy consumption. Additionally, the routing decision time is only 2.4 ms, which, while slightly higher than EASP (1.9 ms) and DSR (1.6 ms), remains well within acceptable bounds and is a reasonable trade-off for the substantial performance improvements gained through intelligent route learning and adaptation. Research evidence shows the ML-powered routing solution has earned worldwide acceptance because of its strong, dependable operation. The model provides a dependable and extended operational outlook for sustainable 6G-enabled wireless sensor networks through its real-time adaptive tracking system using predictive learners.

7. Conclusion

The proposed system, based on machine learning for routing optimization, successfully improved energy efficiency as well as network durability and overall performance within wireless sensor networks that utilize 6G technology. Predictive learning technology within the system operated together with path selection strategies to trigger automatic routing decisions for constantly changing node conditions and link qualities, and network structure modifications. Packet delivery ratio metric and latency performance, and throughput outcome with reduced routing overhead surpassed results from DSR and EASP conventional protocols according to simulation data. Performance validation of the proposed solution included systematic across-metrics and scenario evaluation in addition to quantitative comparisons. Through its implementation, the framework delivered balanced energy consumption alongside failure prevention through a uniform distribution of network traffic. The model demonstrates real-time suitability for large-scale heterogeneous 6 G infrastructure deployments because it offers dynamic condition adaptation through entropy analysis with convergence assessment. The framework can be enhanced as a result of adding reinforcement-based federated learning, which enables private training for dispersed network nodes in a decentralized manner. Research on trust-based routing metrics linked with secure communication protocols needs improvement to strengthen defenses against cybersecurity threats. The combination of energy harvesting technologies and mobile node platforms would build sustainable operations, particularly in adjusting industrial or remote resource areas. New implementations could gain advantages by using quantum communications capabilities that are expected to become available through 6G networks.

Table 5
Comparative performance metrics.

Metric	Proposed model	EASP	DSR
Network Lifetime	1320 rounds	950	870
Avg. Energy (J)	0.52	0.38	0.31
PDR (%)	93.4	81.6	76.2
Delay (ms)	108	136	149
Throughput (pkt/s)	17.6	13.2	11.5
Overhead (pkts)	1420	1830	2100

Table 6
Adaptability and learning metrics.

Evaluation Parameter	Proposed Model	EASP	DSR
Routing Entropy	0.79	0.32	0.28
Convergence Time (s)	7.8	—	—
Avg Hop Count	3.2	4.5	4.8
Routing Decision Time	2.4 ms	1.9	1.6

CRediT authorship contribution statement

Shaymaa Sorour: Writing – review & editing, Writing – original draft, Validation, Methodology. **Nadhem Nemri:** Writing – review & editing, Writing – original draft, Validation, Methodology. **Asmaa Mansour Alghamdi:** Visualization, Validation, Software, Resources, Data curation. **Marwa Obayya:** Writing – review & editing, Writing – original draft, Validation, Methodology, Conceptualization. **Rakan Alanazi:** Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Tawfiq Hasanin:** Writing – original draft, Methodology, Formal analysis. **Noha Alduaiji:** Writing – review & editing, Writing – original draft, Validation. **Alshahrani Saied Falah:** Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.aej.2025.07.032.

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